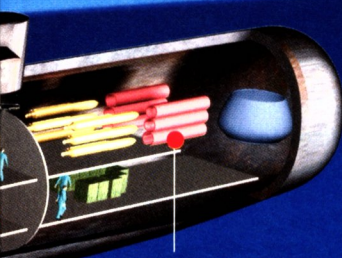


DEEP IMPACT

The historic launch of India's nuclear submarine adds a new member to what was an exclusive Big Five club



Tactical Weapons

Six 533 mm torpedoes/
Klub anti-ship/ land attack
cruise missiles.

Future Missile

A larger variant of the K-15,
the 'K-X', a 3,500-km missile with
multiple warheads is in development.
The *Arihant* will carry four of them.



The sub was built
at SBC, Vizag; it
will be launched
from here.



Site Bravo is
where the sub will
be housed for
harbour trials.



Akula 2 class sub,
being leased from
Russia to train
ATV crews.

N-SUBS OF THE WORLD

	USA	74
	RUSSIA	44
	UK	13
	FRANCE	10
	CHINA	10

GRAPHIC BY PRASHANT CHAUDHARY

cessor. "The launch of the submarine puts us in an exclusive league of five other nations capable of designing and building their own nuclear submarines," says Vice-Admiral (retired) M.K. Roy, the ATV's first project director.

A nuclear submarine is powered by a nuclear reactor which generates tremendous heat that drives a steam turbine. It is, however, one of the most complex machines on earth, the reason why only five countries have the capability. The last country to join was

China, way back in August 1971. Unlike the conventional diesel-electric submarines which have to surface to charge their batteries, nuclear submarines have unlimited underwater endurance and their speed is twice that of their conventional counterparts. Armed with nuclear tipped ballistic missiles (SLBMs), they form the third leg of a nation's nuclear 'triad' comprising air and surface-launched nuclear weapons. Over the next five years, the troika of *Arihant* class SSBNs, each cost-

ing Rs 3,200 crore, will make the third leg of India's nuclear triad—a strategic underwater platform for launching nuclear weapons.

The *Arihant* is, for all practical purposes, a functional, fully-fitted out submarine. After this brief ceremony, the submarine is to be towed out for the first time across the naval dockyard and moored in an enclosed pier called Site Bravo—"from the maternity ward to the nursery", as one official puts it. Over the next few months, it will commence a series of full system harbour trials. The primary system, a nuclear reactor, generates the heat which drives the secondary system, a steam turbine which spins the submarine's propeller, are to be tested separately. First, the steam turbine is to be jump-started with shore-based supply. The next significant step will be starting up the submarine's nuclear reactor where the zirconium rods in the core of the submarine's 80 MW pressurised water reactor will be slowly raised, allowing the reactor to become critical in slow degrees. It will take around three weeks to go fully critical.

Only after all systems are tested, will the primary and secondary systems be mated. If all goes well, the submarine will be allowed to sail out to begin sea trials next year. Weapon trials, including the firing of its arsenal of 12 K-15 short range ballistic missiles, are the last stage of the trials before the submarine is finally commissioned to the navy by 2011. The submarine will carry four of an under-development submarine launched ballistic missile (SLBM) 'K-X' with 3,500-km range, each with several warheads called multiple independently targetable re-entry vehicles (MIRVs). These missiles will enable the submarine to conduct deterrent patrols in proximity to Indian waters. The Indian Navy is only responsible for running and maintaining the nuclear submarines. All its tasking and patrols are to be controlled by the Nuclear Command Authority (NCA) headed by the prime minister. Orders will be passed to the submarine through a secure communication network. The launch of the *Arihant* is a major step forward in India's quest for a minimal but credible nuclear deterrent. Its Asian rival China